

How much does the map of science depend on the embedding model?

Alejandro Treny Ortega

Independent researcher

Manuscript draft • 20 September 2026

Abstract

Science maps use relationships between papers to describe how research is organised. When these relationships come from text embeddings, a model turns each paper into a list of numbers. Would another model give the same picture? I compare ten models using 500,000 OpenAlex publications from 2000–2024 across 26 disciplines, including Medicine and Physics. Representing disciplines by their papers’ average positions produces similar arrangements across models. This holds when other papers are selected to calculate those averages. Yet within a discipline and period, two models share, on average, only eight of the 25 papers closest to a given paper. Giving a model each paper’s title alone, instead of its title and abstract, also changes much of that list. The effect varies between models. One model can place a discipline’s papers closer together than another’s, while another model reverses that order. This comparison uses angles between paper positions. Moving their reference point can change the answer even when selecting other papers does not. These checks show which relationships are shared across representations and which depend on the model, input text or reference point. Agreement about the broad arrangement can coexist with different answers about nearby papers and comparisons between disciplines.

Keywords: science mapping; document embeddings; robustness; neighbourhood agreement; geometric comparisons.

1 Introduction

A science map can help a reader find related papers or compare research areas. Its usefulness depends on the relationships it shows. If two models place different papers close to the same article, the reader may follow different routes through the literature. If they disagree about how widely an area’s papers are spread, they may also support different descriptions of that area. I ask how much of this changes when the papers stay the same and only their representation changes.

Text embeddings provide one way to build these maps. A model reads a paper’s text and turns it into a list of numbers, called a vector. Those numbers give the paper a position

in a space where distances can be calculated. Papers placed close together are treated as related. This approach has been used to map biomedical literature and the structure of physics (González-Márquez et al., 2024; Constantino et al., 2025). But the position comes from the model. It is not a relationship observed independently of that model.

This concern has a history in science mapping. Gläser et al. (2017) asked how much a reconstructed thematic structure reflects scientific knowledge and how much it reflects the method used to recover it. Boyack and Klavans (2020) found that methods could obtain similar overall accuracy while assigning different papers to clusters. An overall score can therefore leave an important question unanswered: would another method support the particular comparison a reader wants to make?

I address that question using ten models and a fixed collection of 500,000 publications. First, I compare the broad arrangement of research areas and the articles placed near each other. I then ask whether changing the text supplied to a model matters as much as changing the model itself. A later, exploratory analysis examines specific comparisons between areas: does each model identify the same area as having more widely spread paper representations? Repeating these comparisons with other articles helps distinguish dependence on the selected papers from dependence on the model.

2 Background and research questions

2.1 What earlier comparisons tell us

Previous studies assess scientific embeddings against several useful criteria. Constantino et al. (2025) compare them with disciplinary classifications in physics; González-Márquez et al. (2024) examine related biomedical papers and their labels. Such checks ask whether a representation agrees with information about the literature. The present study asks a complementary question: which relationships remain when the representation changes?

The model is only one choice involved. A title and a full abstract contain different amounts of information. Models also read different lengths of text and combine their outputs in different ways. Singh and Singh (2022) show that scholarly search can be sensitive to the wording of short queries. This makes the text supplied to a model part of the comparison, rather than a detail to leave unspecified.

Comparing both overall structure and nearby papers is not new. Caspari et al. (2024) combine geometric and neighbourhood comparisons, while Imel and Hafen (2025) examine properties of scientific embedding spaces alongside citation relationships. I build on these approaches by keeping articles matched between models, changing text and processing choices explicitly, and checking specific comparisons between research areas with repeated article selections and alternative measures.

2.2 What agreement can establish

Throughout the article, an *article selection* means choosing a set of papers from the fixed collection. Repeating the selection means choosing another set under the same size and date rules, without adding newly downloaded papers. *Agreement* asks whether different models give similar results for the same papers. *Selection stability* asks whether a result repeats when the selected papers change. A comparison can be stable within each model while the models give opposite answers.

Neither property establishes that the answer is correct about the research itself. Models can share an error, and a repeatable pattern can be hard to interpret. In a study of invasion biology, Held and Velden (2022) show why topic solutions require substantive interpretation beyond numerical checks. Here, agreement is evidence that a result survives a change of model, not evidence of thematic truth.

I compare the numerical representations before they are reduced to a drawing on a page. Each vector has hundreds of coordinates; a two-dimensional plot would add another transformation that could change apparent distances. In this article, the shape of a map refers to relationships in that original space.

2.3 Research questions

RQ1. When the model changes, what remains of the arrangement of research areas and of the papers placed close together?

RQ2. How much does changing the input text, or the rule used to form one paper vector, matter compared with changing the model?

RQ3. Do models agree about which research area has more widely spread paper representations, or variation across more directions? Do these comparisons repeat with other articles?

The third question was added after the initial comparisons and is exploratory. All three concern a fixed collection and a fixed set of models. They do not seek a winning model or estimate the structure of all scientific literature.

3 Data and methods

3.1 One collection, several questions

The starting collection contains 500,000 distinct OpenAlex work IDs, retrieved and cleaned by 15 September 2026 (Priem et al., 2022). Records were labelled as English articles, reviews or conference papers published in 2000–2024. Each required a title, a reconstructed abstract of at least 50 words and a primary research field. Retractions, paratext and clear content failures were excluded. Less certain cases and possible duplicate versions were kept with flags, rather than treated as resolved. Supplement S1 gives the rules and counts.

OpenAlex assigns papers to broad *Fields* and narrower *Subfields*, using information from text, citations and journals (OpenAlex, 2026). I call these areas and specialties. For example, Medicine is an area; Cardiology and Cardiovascular Medicine is one of its specialties. The 26 areas provide the main units of comparison. These database labels organise the analysis; they are not independently verified descriptions of every paper.

The collection has two parts. Seeded random API requests supplied 400,000 eligible records. Another 100,000 strengthened coverage where few papers were available within an area and five-year period. The five periods are 2000–04, 2005–09, 2010–14, 2015–19 and 2020–24, giving 130 area–period groups with at least 2,165 papers each. Global searches use the 400,000-record general collection. Area comparisons use both parts. Neither the reinforced collection nor an average giving every area equal weight estimates world publication proportions.

Not every question requires processing all 500,000 papers again. To compare titles, abstracts and both together, I use the same 52,000 papers: 2,000 per area, with 400 from each period. This includes the 26,000 papers used in an earlier, smaller trial of that same comparison. The later analysis of spread within areas reuses these 52,000 papers. A separate set of 52,000 serves a different purpose: checking what happens when every model reads the same passage of text. Table 1 keeps these uses apart. In every model comparison, both models receive the same article IDs.

Table 1: Questions, comparisons and article sets.

Question	What is compared	Articles used
Does the broad arrangement stay similar?	Ten models: relations among area or specialty centres, and among individual papers inside groups.	256 papers per centre; repeated choices and size checks at 128 and 512.
Do models find the same nearby papers?	The overlap between two models' lists of 25 neighbours.	All papers within each area and period; a separate check fixes 2,048 candidates per group. Global search: 13,000 starting papers among 400,000 candidates.
Does changing the input text matter?	The same model reads only the title, only the abstract, or both.	52,000: 2,000 per area, 400 per period. Includes the first 26,000-paper trial.
Does reading the same passage matter?	Each model's usual input versus a common passage that fits every model.	A different set of 52,000, with 400 per area and period.
Which area has more spread or more directions of variation?	Two geometric properties for all 325 pairs of areas, under ten models.	The same 52,000 as in the input-text experiment; further article choices test whether the comparisons repeat.
Does searching a narrower specialty matter?	For example, Cardiology and Cardiovascular Medicine versus Medicine, with 256 candidates in each search.	50 fixed starting papers per specialty; ten choices of the other 206 candidates. In total, 10,850 starting papers across 217 specialties.

All sets come from the same 500,000-paper collection: 400,000 general records plus 100,000 added to strengthen small area–period groups. Rows are not added together. Models use the same paper IDs within each comparison. In the scope comparison, both candidate sets include the same 50 starting papers. Methods and Supplements S1–S4 and S9 give the full controls. The 1,300 software-test papers are not another scientific sample.

3.2 How each model represents a paper

The ten models in Table 2 cover scientific documents, general sentence similarity, and scientific or biomedical language. They were selected for their use in scholarly representation and their different training approaches. They are a fixed comparison set, not a random sample of models or a ranking of the newest systems. Model versions, text formats and software settings were fixed before their representations were calculated.

The main input is the title followed by the abstract, in each model’s usual format. Models read small pieces of text called *tokens*, which may be whole words or parts of words. Their input limits therefore differ from word counts. MiniLM reads at most 256 tokens and MPNet 384 in their standard settings. These limits cut off part of the input for 49.58% and 20.16% of papers, respectively. I retain those settings and test MiniLM at 512 tokens separately. The common-passage check described above lets every model read the same content, although each divides it into tokens in its own way.

Some models produce a vector for each token, so a further rule must turn those outputs into one vector for the paper. This step is called *pooling*. For BERT, SciBERT, BioBERT and PubMedBERT, the main rule averages the last-layer outputs at all positions that contain text or special symbols, excluding padding. These four models were not trained specifically to compare whole documents. Two checks instead use the first-position output, called CLS, or the last non-padding output, called SEP. CLS is taken before the model’s additional pooler transformation. The mean rule was specified before the primary comparisons and retained after early checks; it was not chosen to maximise agreement. The other six models use their documented rules. Supplement S1 provides exact settings.

3.3 Following the map from areas to individual papers

Before comparing representations, each article vector is scaled to length one. This makes comparisons depend on direction rather than vector length. To summarise an area, I average its article vectors and scale that centre to length one as well. The arrangement of these centres describes broad relationships between areas. Comparisons of the individual vectors inside each area retain the detail hidden by that average.

The original centre comparison uses 256 articles per centre: 26 areas, 217 specialties, or one specialty from each area. I then repeat article selection fifty times, keeping each group’s period counts. A size check uses 128, 256 and 512 papers, with the same 26 areas and 183 eligible specialties. Each selection has a random-group reference made from exactly the same articles and date counts. Further checks omit each area and separate the choice of specialty from the choice of its papers. These centre comparisons are distinct from comparing individual vectors within groups (Supplement S9).

Table 2: Embedding models and primary representation settings.

Model	Vector size	Token limit	Paper vector	Reference
SPECTER	768	512	CLS	(Cohan et al., 2020)
SPECTER2	768	512	CLS + adapter	(Singh et al., 2023)
SciNCL	768	512	CLS	(Ostendorff et al., 2022)
SciBERT	768	512	Mean	(Beltagy et al., 2019)
BERT	768	512	Mean	(Devlin et al., 2019)
MPNet	768	384	Mean	(Reimers and Gurevych, 2019)
MiniLM	384	256	Mean	(Reimers and Gurevych, 2019)
PubMedBERT / BiomedBERT	768	512	Mean	(Gu et al., 2022)
BioBERT	768	512	Mean	(Lee et al., 2020)
SimCSE	768	512	CLS, pre-pooler	(Gao et al., 2021)

Vector size is the number of coordinates. MPNet and MiniLM use all-mpnet-base-v2 and all-MiniLM-L6-v2, respectively. Limits count tokens including special symbols. Mean: last-layer average over non-padding positions, including special symbols. CLS: first-position state before the pooler; SPECTER2 adds its proximity adapter. For MPNet/MiniLM, the reference describes the Sentence-BERT framework, not the exact checkpoint training. Changing the pooling rule does not add a separately trained model. Full revisions, formats and truncation: Table S2.

For an individual paper, its *neighbours* are the papers placed closest to it. The *starting paper* is the one whose neighbours we seek; the *candidates* are the papers available for that search. I compare the lists of 25 nearest neighbours produced by two models, with lists of 10 and 50 as checks. Closeness is measured by cosine similarity, which compares vector directions. The starting paper cannot be its own neighbour. Lists are calculated exactly and checked by independent sorting. Each result specifies whether candidates come from the whole area and period, an equal-size selection within that group, or the 400,000-record general collection.

The quality check applies the existing flags and duplicate rule, leaving 448,886 records. It compares the same retained starting papers before and after filtering candidates. Equal-size controls use 1,024 candidates in all 130 groups and 2,048 in the 63 groups large enough after filtering. They include the same 100 clean starting papers in both candidate sets. A separate text check matches dates and candidate counts inside the 52,000-paper experiment. These controls test the recorded filter, not every possible source error.

A further comparison asks whether models share more neighbours when the search stays within a specialty or extends to its parent area. Consider 50 fixed starting papers from Cardiology and Cardiovascular Medicine. For each, we compare searches among 256 cardi-

ology papers and among 256 papers from Medicine. Both candidate sets include those same 50 starting papers, leaving 206 places for other papers; each search excludes the starting paper itself. I repeat the choice of the other 206 ten times, keeping the 50 starting papers fixed and matching publication-period counts between searches. This procedure covers 217 specialties, giving 10,850 starting papers. The broader set is therefore deliberately anchored to the specialty’s 50 papers, rather than freely sampled from the whole area.

3.4 Measuring shared structure and spread within an area

To compare overall structure, I ask whether the same papers have similar patterns of relationships in two representations. The main measure is corrected linear centred kernel alignment, abbreviated CKA (Kornblith et al., 2019; Murphy et al., 2024). A value near one, such as 0.9, indicates strong similarity between these patterns on this measure: the models organise the same objects in similar ways. It is an agreement score, not a percentage of correctly represented science. The correction reduces a known inflation caused by finite samples; it does not remove all statistical bias. Two further checks allow a rotation and rescaling of the representations (Procrustes alignment), or compare the ranking of distances between papers. They respond to different aspects of the representations (Williams et al., 2021; Williams, 2024). Supplement S2 gives the formulas.

I also describe two properties within each area. *Angular spread* measures how far apart article vectors point: it summarises the angles between distinct vectors, using the median-distance convention in Supplement S6. Larger angles mean a wider spread of directions. *Effective linear dimension*, measured by the participation ratio (PR), asks how widely variation is shared among directions. For intuition, a cloud concentrated along one line varies mainly in one direction; a cloud spread across a plane varies in more. The measure extends that idea to the full vector space. Neither property counts research topics or establishes how diverse an area is.

Alternative angle summaries and measures of variation test dependence on the chosen formula. For dimension, one alternative measures how evenly variation is distributed (entropy-based dimension); another counts the directions needed to account for 80% of it (D80). These alternatives use the same underlying variation, so they do not independently validate a thematic interpretation. I also tested whether links between nearby papers could identify separate groups, but checks with artificial examples prevented treating that measure as general fragmentation. Definitions and controls appear in Supplement S6.

3.5 Checking whether a comparison repeats

With 26 areas, there are 325 ways to choose two different areas, counting each comparison once. For each geometric property, I ask which area has the larger value under each model. I repeat the calculation using twenty half samples of 1,000 papers per area and five further selections of 2,000. A direction is called persistent only if it holds in all these selections and in the main set of 2,000: 26 conditions in total. The two dimension alternatives use the same conditions. A separate check changes the point from which vector directions are measured. For each model, I calculate the mean of the balanced 52,000 unit vectors, subtract this common mean from every vector and restore each vector to length one. This operation, called global centring followed by normalisation, moves the reference point to the collection's mean position. It changes the representation before measuring spread and dimension; it is distinct from the covariance centring within PR.

This gives three outcomes: all ten models consistently identify the same area as having the larger value; at least two fixed models consistently give opposite answers; or neither condition holds. I call the third outcome *unresolved*. For example, some models may change their ordering when other papers are selected, without any fixed pair maintaining opposite answers. A tied or changing ordering does not count as a persistent direction. If neither complete agreement nor fixed opposition remains, the pair is unresolved; this does not mean the areas are equal. Besides any nonzero difference, I examine minimum differences of 1, 5 and 10% of the two values' average. A difference below the required minimum also remains unresolved. These cutoffs show whether very small differences drive a result.

Two selections can contain many of the same papers because both come from the fixed collection. The range of their results therefore describes what changes when we choose other available papers; it is not a confidence interval for all scientific literature. A *selection alert* marks a result whose variation across article choices, or shift between tested sizes, exceeds a recorded tolerance. It identifies sensitivity to the selected articles, not a source-record error. The supplement gives each check's rules and counts. Comparisons also reuse models and areas, so their results are not treated as independent observations in significance tests.

The work developed in stages. The first structure and neighbour comparisons preceded the expanded text experiment and the specialty searches. The geometric analysis was a later exploration; the 325-comparison summary followed its initial examples. Rules were recorded before the corresponding new calculations, with earlier results already known. Some controls were added during this process. Supplement S2 records the sequence; the study was not externally preregistered as a whole.

4 Results

4.1 The broad arrangement is shared more than the detail

The broad arrangement repeats when different papers are selected. The original mean corrected CKA values of 0.939 for area centres and 0.911 for specialty centres become 0.940 and 0.911 across fifty selections. Their central 95% selection ranges are 0.937–0.943 and 0.910–0.912 (Figure 1A). Random groups average 0.627 and 0.636; none of their selection means reaches the lowest observed selection mean.

The size check gives area-centre means of 0.938, 0.940 and 0.941 at 128, 256 and 512 papers; the same 183 specialties give 0.913, 0.914 and 0.915. Omitting one area from the original comparison gives 0.933–0.946. No single area accounts for the high overall agreement. Choosing one specialty per area gives a mean of 0.904: which specialty is chosen matters more than which papers are chosen within it (Table S18).

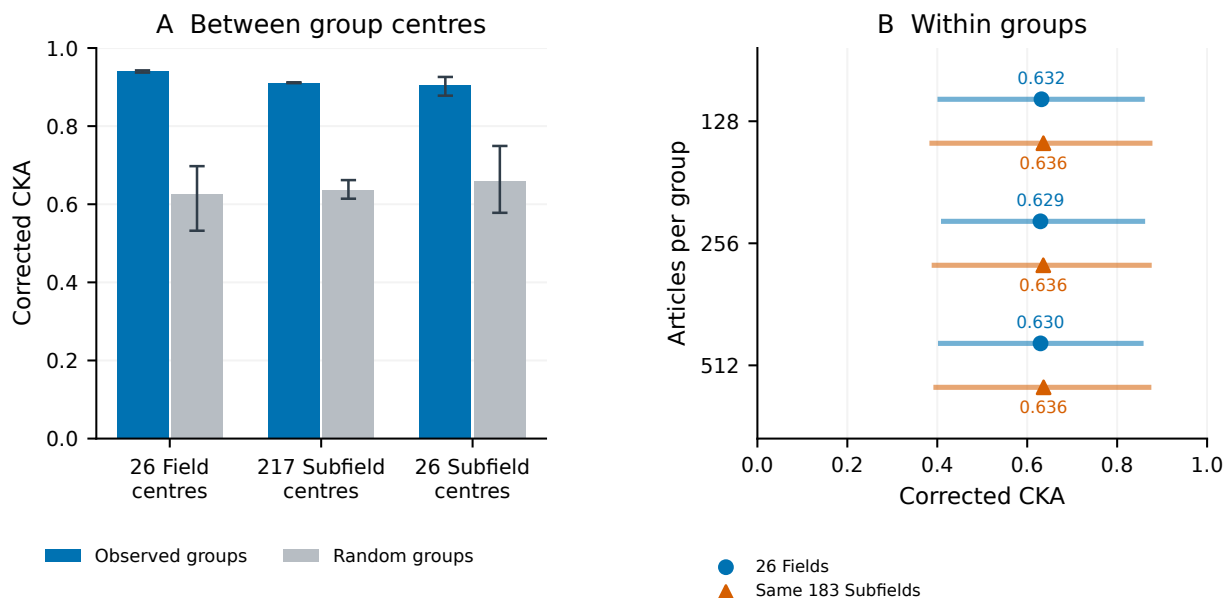


Figure 1: Agreement about area summaries and the papers inside them. Higher corrected CKA means greater agreement between models. (A) Centres use 256 papers each. Area and specialty results average fifty article selections. The 26-specialty condition instead chooses one specialty from each of the 26 areas, repeats that choice fifty times, and chooses ten sets of articles for each choice. Random groups use the same articles, sizes and dates. Whiskers show 2.5th–97.5th percentiles of selection means. (B) Agreement among individual papers at three article counts, within 26 areas and the same 183 specialties. Points show means; lines span the 2.5th–97.5th percentiles of group-model-pair results: 1,170 for areas and 8,235 for specialties. The panels summarise different objects and sources of variation; these ranges are not confidence intervals.

Inside groups, mean CKA remains around 0.63 for areas and 0.64 for the same 183 specialties from 128 to 512 articles (Figure 1B). This stable average hides sensitive local comparisons: 31.9% of the 8,235 specialty–model-pair comparisons trigger the selection alert (2,628 cases). Their agreement varies more than the recorded tolerance allows when the articles or tested sample size change. Thus, a stable overall average can coexist with specialty results that depend on which papers are chosen. The difference between centres and individual papers also compares different objects, rather than isolating scale alone.

The alternative measures broadly support the ordering of model agreements. Across the original 130 area–period groups, their mean rank correlations with corrected CKA are 0.891 for Procrustes and 0.881 for distance ranks. Agreement is not complete, and the original selection check retains 50 alerts among 5,850 model-pair/group comparisons (Table S4). These particular cases remain relevant even when the overall mean changes little.

4.2 Changing the model changes which papers appear nearby

Within the same area and period, two models retain an average of 30.3% of the 25 nearest neighbours: about eight papers out of 25. This average gives equal weight to each group and model pair. Limiting every area–period group to 2,048 candidates raises agreement to 31.9%. The count is the same across groups; the article IDs are shared across models within each group, not across different disciplines. A separate search takes 100 starting papers from each of the 130 groups, giving 13,000 in total, and finds their neighbours among the 400,000 general records. Here, agreement is 17.5%. That search both admits papers from other disciplines and periods and offers many more candidates. Either change could affect agreement, so their effects cannot be separated by this comparison.

I then remove records carrying the tested source flags and keep one representative of each detected duplicate group. For the same retained starting papers, neighbour agreement changes from 30.58% to 31.43%: a gain of 0.85 percentage points. Both averages still amount to roughly eight shared papers in a list of 25. Holding candidate counts fixed makes the change smaller: 0.28 points with 1,024 candidates and 0.19 with 2,048. The issues removed by this filter therefore explain little of the overall disagreement, although some areas change more than others (Table S19).

The paired searches let us change the disciplinary scope while keeping candidate counts and dates matched (Figure 2). Searching only within a specialty makes models share fewer of their 25 nearest neighbours in 127 of 217 specialties (58.5%), but more in the other 90 (41.5%). The average change relative to searching the parent area is -1.35 percentage points. With lists of 50 neighbours, the split is almost even: 109 specialties decrease and 108 increase. Restricting the search therefore does not have the same effect everywhere; these changes concern overlap between models’ lists, rather than how useful the listed papers are.

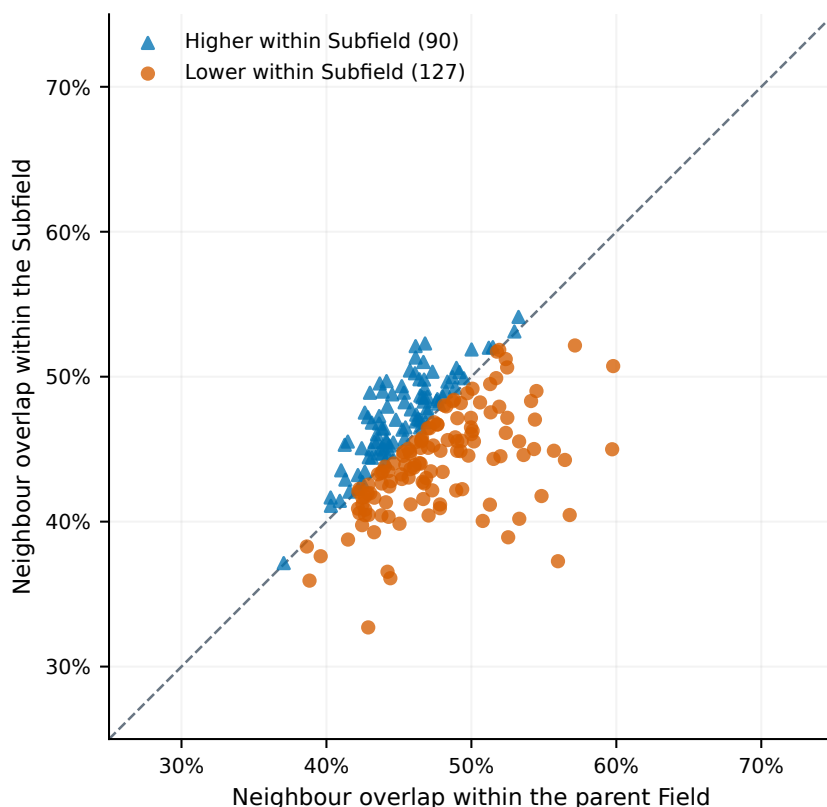


Figure 2: Does restricting a search to a specialty change agreement? Each point represents one of 217 specialties. The horizontal axis shows the fraction of shared neighbours when searching the parent area, such as Medicine; the vertical axis shows that fraction within its narrower specialty, such as Cardiology and Cardiovascular Medicine. A point above the diagonal means that models share more neighbours in the narrower search. Both candidate sets contain 256 papers, including the same 50 starting papers, and matched dates. Values average model pairs and ten choices of the other 206 candidates. The parent-area set therefore always includes those 50 specialty papers. The spread of points shows differences between specialties, not confidence intervals.

Overall structure and neighbour agreement are related: their rank correlation across the 45 model-pair averages is 0.920. In other words, model pairs that agree more on one tend to agree more on the other. Yet a pair can preserve broad structure while changing many individual neighbours. These shared-model observations are descriptive, not independent evidence of reliability.

Supplement S4 moves from averages to named papers: it checks whether a paper keeps the same neighbours when sample size, the rule for forming its vector, or neighbour-list length changes. Reading these examples also revealed source problems. A history book appears under Physics, a text about bacterial infection under Politics, and another record is an announcement. I checked these against the downloaded OpenAlex records: the classifications and content were already present there, rather than introduced by matching the wrong papers

across models. The examples remain in the analysis. Because they were selected as cases to inspect, their number does not estimate the frequency of such problems in the collection.

4.3 Similar average changes hide different model behaviour

For each of the same 52,000 papers, every model receives three inputs in turn: title only, abstract only, and title plus abstract. The combined text provides the starting representation. I then compare two changes: keeping that text and using another model, or keeping the model and supplying only the title. On average, the first replaces 68.8% of the 25 neighbours and the second replaces 70.7% (Figure 3). The corresponding changes in overall structure, measured as one minus CKA, are 0.370 and 0.366. These are averages over the model panel, not the effect for every model.

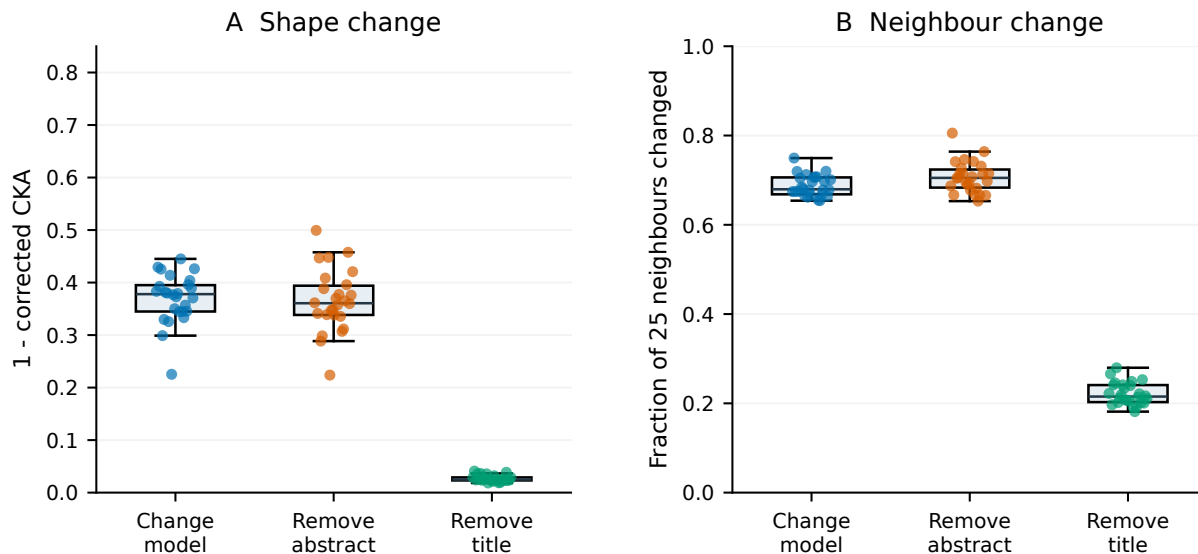


Figure 3: Changing the model or changing what it reads. All comparisons use the same 52,000 papers, with 2,000 per area. The reference input is title plus abstract. Model change compares different models on that input; text change keeps the model and uses only the title or only the abstract. Larger values mean more change: (A) one minus corrected CKA; (B) the fraction of 25 neighbours replaced. Each point is an area average. Boxes show the median and middle half of area values; whiskers extend to values within 1.5 interquartile ranges of the box. These are descriptive distributions, not confidence intervals. Models retain their usual text limits.

Using only the abstract changes less: 0.027 for structure and 22.1% of neighbours. The averages also hide differences between models. Title-only input replaces 51.7% of neighbours for MPNet and 84.2% for SciBERT (Table S5).

A direct neighbour check repeats fifty balanced selections of 1,000 candidates per area, half the full panel. At 25 neighbours, model change replaces 64.93% and title-only input replaces 66.49%. The difference is 1.56 percentage points, with a central 95% selection range of 1.44–1.69; its sign never reverses. The mean differences are 1.97 points at 10 neighbours and 1.04 at 50. Yet model-specific differences range from -14.04 points for MPNet to $+16.59$ for SciBERT. The stable grand mean conceals opposing model-specific results (Supplement S9).

The result also depends on how BERT, SciBERT, BioBERT and PubMedBERT produce one vector for a whole paper. The main rule averages their token outputs. The alternatives use just the first output (CLS) or the last non-padding output (SEP), each of which has processed the input context. With these alternatives, title-only input changes fewer neighbours than switching models in every repetition: 3.13 percentage points fewer for CLS and 2.00 fewer for SEP at 25 neighbours. Both comparisons are recalculated with the chosen rule; the other six models retain their usual settings. Thus, the average ordering reverses when we change this practical way of using the four models.

This three-input experiment began with 26,000 papers and then added another 26,000 under the same design; the first set is included in the final 52,000. The same structural selection check was applied at both sizes, choosing other papers 100 times. Alerts fall from 72 of 520 model–area–input comparisons (13.8%) to five of 520 (1.0%). Fewer results therefore exceed the recorded tolerance, although this does not establish stability for every neighbour comparison. Supplement S3 keeps the earlier twenty-selection check separate. The common-passage and MiniLM 512-token checks address a different question, input length; they do not reveal a coding error in MiniLM’s original limit.

4.4 A comparison between two areas can reverse under another model

The geometric analysis reuses the 52,000-paper text sample and its title-plus-abstract representations. It asks whether, for example, one area’s vectors point further apart than another’s. I first checked how much each measure changes when different articles are chosen. Across ten models and 26 areas, angular spread has no alerts in the half-sample or equal-size checks. PR has four alerts among 260 cases (1.5%) when using twenty half samples of 1,000 papers per area. None appears in five further selections of 2,000 papers per area. These checks use different sizes and repetition counts; passing the second does not erase sensitivity found in the first.

Of the 325 area pairs, 80.6% have persistent opposite answers from at least two models for angular spread; the share is 68.0% for PR (Figure 4). For example, one fixed model consistently gives area A a larger angular spread than B, while another consistently gives B the larger spread. “Larger” refers to the measured value, not a higher position in a drawing or better research. Other models need not disagree with one another. All ten consistently agree

on the larger area in 12.9% of pairs for angular spread and 16.6% for PR. The remaining 6.5% and 15.4%, respectively, do not meet either rule and remain unresolved.

Many opposite answers involve small differences. Requiring a difference above 5% of the two area values' average reduces persistent opposition to 29.5% of all 325 pairs for angular spread and 54.5% for PR. At a 10% minimum difference, these shares fall to 5.8% and 38.8%. A reversal involving a tiny difference should therefore be distinguished from one involving a larger difference.

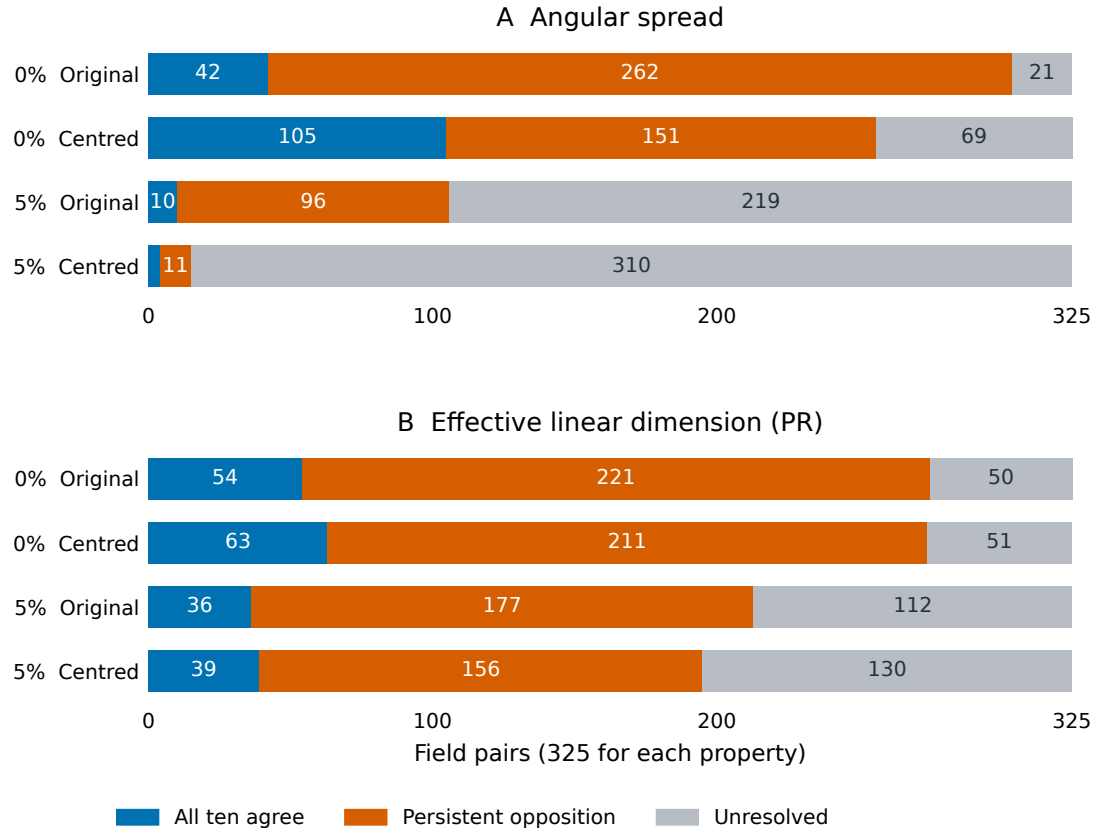


Figure 4: The answer depends on the representation as well as the model. Each bar covers all 325 area pairs. Blue: all ten models consistently give the same area the larger measured value. Orange: at least two fixed models consistently give opposite orderings. Grey: neither rule is met, including ties and changing orderings. Every classification uses 26 conditions: the main 2,000 papers per area, twenty half samples of 1,000, and five further selections of 2,000. The original and centred representations use identical papers. Centring subtracts each model's mean over the balanced 52,000 unit vectors, then rescales each resulting vector to length one. Cutoffs of 0 and 5% require relative differences above those values in every condition. Table S20 gives all four cutoffs and tracks whether the same opposing models retain their directions.

Alternative angle summaries retain at least one of the original opposing model pairs and its directions in 85.9% of the original angular oppositions (225/262). For dimension, the entropy measure summarises how evenly variation is distributed across directions; D80 counts the

fewest directions needed to explain 80% of that variation. Across the same 26 conditions, entropy retains such an opposing pair in 94.6% of the original PR oppositions and D80 in 86.0%; the same 86.0% survive both alternatives (190/221). D80 can tie because it returns whole numbers, leaving those comparisons unresolved. These checks show how much the ordering depends on the formula. Restricting the panel to the six similarity-trained models still leaves opposite answers in 71.1% of all 325 pairs for angular spread and 49.2% for PR (Tables S14–S15).

Moving the reference point to the collection’s mean position changes many angular comparisons (Figure 4). After subtracting this mean and restoring unit vector length, persistent opposition falls from 80.6% to 46.5% of all 325 pairs (262 to 151). Fewer than half of the original oppositions, 48.9% (128/262), retain at least one of the same opposing model pairs in the same directions; Table S20 distinguishes these from new or changed oppositions. Requiring differences above 5% leaves only 11 opposed pairs instead of 96. PR changes less: opposed pairs fall from 221 to 211, and 92.3% of its original oppositions retain at least one of the same opposing model pairs and its directions (204/221). Thus, an angular comparison that repeats with other papers can still depend strongly on where the shared reference point is placed.

I also asked whether the network linking nearby papers could reveal an area split into separate groups. This would matter if weak connections could be read as a division into research topics. The checks did not support that reading: artificial elongated clouds and isolated extreme points can produce weak-connection signals without any thematic groups. The signal also changes with sample size and the rule for linking neighbours, with selection alerts in 51 of 260 cases. Supplement S6 retains this unsuccessful check so that weak geometric connection is not mistaken for evidence of thematic fragmentation.

5 Discussion

5.1 What should be checked before using a map

The models share a broad arrangement, yet can disagree about nearby papers or comparisons between areas. Suppose one model gives area A a larger angular spread than B, and this repeats with other papers. That establishes stability to article selection within that model. If another model consistently reverses the order, repeating the first model’s selections cannot settle the disagreement. A map-based claim therefore needs to be checked by changing the model as well as the articles.

The concern follows earlier work on method-dependent science maps (Gläser et al., 2017; Boyack and Klavans, 2020) and complements assessments against external classifications and relationships (González-Márquez et al., 2024; Constantino et al., 2025). Combining structural and neighbour comparisons also has precedent (Caspari et al., 2024). The contribution

here is the controlled comparison across levels, text choices and repeated area comparisons, including cases where a result persists within models but reverses between them.

Interpretation still depends on what the measure describes. A larger angle means that vectors point further apart. A larger effective linear dimension means that variation is less concentrated in a few directions. Neither, by itself, shows greater thematic diversity or interdisciplinarity. Related work on embedding geometry also cautions against assigning a broad meaning to a convenient measure (Rudman et al., 2022). The failed fragmentation check makes this concrete: a repeatable difference can still support the wrong explanation.

5.2 What the additional checks change

The checks help separate findings that repeat from choices that still affect them. Recalculating area averages with other papers, with 128, 256 or 512 papers, or leaving out one area at a time preserves high agreement about the broad arrangement. Removing the flagged and duplicate records barely changes overall neighbour overlap: the two models still share about eight papers in a list of 25.

For input text, repeating article selection preserves the average comparison, but that average combines different model behaviours. Giving only titles changes fewer neighbours than switching models for some models, and more for others. Replacing the averaging rule with CLS or SEP for the four BERT models also reverses the overall ordering. The practical implication is to specify both what text a model reads and how its outputs become a paper vector; the average cannot stand for every configuration.

For angular spread, the reference point is another consequential choice. The same papers can give a different answer about which area is more widely spread when angles are measured from the collection’s mean position and vectors are rescaled. Repeating article selection does not remove that dependence. PR changes less under this transformation, but its alternative formulas do not retain every comparison either. A claim that one area is more spread out should therefore identify the representation and measure used, rather than leave these choices implicit.

The discipline checks ask why Medicine and Energy differ (Supplement S5). With the main rule, models agree less on relationships among Medicine papers than among Energy papers, although Medicine does not always have the lowest neighbour overlap. The difference remains after balancing the representation of specialties within each area and after excluding BioBERT and PubMedBERT. These checks therefore do not identify a single explanation. Models with related training also sometimes give similar results, but these apparent families change when the rule for forming paper vectors changes. With only ten models that differ in several ways at once, the comparisons cannot isolate training data, architecture or training objective as the cause.

The time comparison uses the same current models on papers from older and newer publication periods. Structural agreement increases slightly on average. Neighbour agreement falls by 0.87 percentage points when each group offers all its available papers, but rises by 0.91 points when every group offers 2,048 candidates. Older and newer groups contain different numbers of papers; fixing that number reverses the direction of the result. This shows why candidate counts must be controlled before interpreting a time trend. It does not establish that science itself has historically converged.

5.3 What remains open

The collection requires English-labelled records with usable abstracts. Its reinforcement of smaller area–period groups helps comparison but changes proportions. Source errors remain, and cleaning is not a certification of every record. Repeated selections cannot repair missing coverage or make these results representative of all scientific work.

The ten models also set a boundary. Four were not trained specifically for document similarity. The six-model check helps show that disagreement is not confined to those four, but does not make the panel representative of all encoders. Deciding which direction is better supported requires evidence outside the embeddings, such as independently assessed article relationships.

Finally, several extensions and controls followed earlier results. Recording their rules before new calculations improves traceability without turning the whole study into a confirmatory design. A useful next step would test a defined map-based claim against independent judgments in a research area. Adding models alone would not settle what their shared or conflicting geometry means.

6 Conclusion

Changing the embedding model can preserve the broad arrangement of science while changing which papers appear nearby and which area has more widely spread representations. Giving a model only titles can also produce large changes, whose size depends on the model and the rule for forming paper vectors. Repeating the analysis with other papers is useful, but does not show that another model or reference point would support the same answer. Before using a map to describe a relationship, check that relationship under those choices and report how large the changes are. These comparisons identify what the tested representations share; judging whether the relationship describes the research well still needs independent evidence.

Declarations

Author contributions. Alejandro Treny Ortega conceived the project and directed the research and manuscript preparation as an independent researcher.

Competing interests. The author declares no competing interests.

Funding. This research received no external funding.

Data and code availability. The project repository is <https://github.com/aleetreny/Shape-of-Science-Robustness>. This draft is accompanied by the tables used for its figures and the editable manuscript sources. The frozen corpus, vectors and complete local execution records have not been deposited in a persistent public archive. Public code alone does not provide full reproduction of the present analysis. The access inventory and remaining release requirements are stated in Supplement S8; no data deposit or persistent identifier is claimed here.

References

- Iz Beltagy, Kyle Lo, and Arman Cohan. SciBERT: A pretrained language model for scientific text. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3615–3620, Hong Kong, China, nov 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1371. URL <https://aclanthology.org/D19-1371/>.
- Kevin W. Boyack and Richard Klavans. A comparison of large-scale science models based on textual, direct citation and hybrid relatedness. *Quantitative Science Studies*, 1(4):1570–1585, 2020. ISSN 2641-3337. doi: 10.1162/qss_a_00085.
- Laura Caspari, Kanishka Ghosh Dastidar, Saber Zerhouni, Jelena Mitrovic, and Michael Granitzer. Beyond benchmarks: Evaluating embedding model similarity for retrieval augmented generation systems. arXiv:2407.08275, 2024. URL <https://arxiv.org/abs/2407.08275>. Preprint.
- Arman Cohan, Sergey Feldman, Iz Beltagy, Doug Downey, and Daniel Weld. SPECTER: Document-level representation learning using citation-informed transformers. In Dan Jurafsky, Joyce Chai, Natalie Schluter, and Joel Tetreault, editors, *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, pages 2270–2282, Online, jul 2020. Association for Computational Linguistics. doi: 10.18653/v1/2020.acl-main.207. URL <https://aclanthology.org/2020.acl-main.207/>.
- Isabel Constantino, Sadamori Kojaku, Santo Fortunato, and Yong-Yeol Ahn. Representing the disciplinary structure of physics: A comparative evaluation of graph and text embedding methods. *Quantitative Science Studies*, 6:263–280, 2025. ISSN 2641-3337. doi: 10.1162/qss_a_00349.

- Jacob Devlin, Ming-Wei Chang, Kenton Lee, and Kristina Toutanova. BERT: Pre-training of deep bidirectional transformers for language understanding. In Jill Burstein, Christy Doran, and Thamar Solorio, editors, *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies, Volume 1 (Long and Short Papers)*, pages 4171–4186, Minneapolis, Minnesota, jun 2019. Association for Computational Linguistics. doi: 10.18653/v1/N19-1423. URL <https://aclanthology.org/N19-1423/>.
- Tianyu Gao, Xingcheng Yao, and Danqi Chen. SimCSE: Simple contrastive learning of sentence embeddings. In Marie-Francine Moens, Xuanjing Huang, Lucia Specia, and Scott Wen-tau Yih, editors, *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 6894–6910, Online and Punta Cana, Dominican Republic, nov 2021. Association for Computational Linguistics. doi: 10.18653/v1/2021.emnlp-main.552. URL <https://aclanthology.org/2021.emnlp-main.552/>.
- Jochen Gläser, Wolfgang Glänzel, and Andrea Scharnhorst. Same data—different results? towards a comparative approach to the identification of thematic structures in science. *Scientometrics*, 111(2):981–998, 2017. ISSN 0138-9130. doi: 10.1007/s11192-017-2296-z.
- Rita González-Márquez, Luca Schmidt, Benjamin M. Schmidt, Philipp Berens, and Dmitry Kobak. The landscape of biomedical research. *Patterns*, 5(6):100968, 2024. ISSN 2666-3899. doi: 10.1016/j.patter.2024.100968.
- Yu Gu, Robert Tinn, Hao Cheng, Michael Lucas, Naoto Usuyama, Xiaodong Liu, Tristan Naumann, Jianfeng Gao, and Hoifung Poon. Domain-specific language model pretraining for biomedical natural language processing. *ACM Transactions on Computing for Healthcare*, 3(1):1–23, 2022. ISSN 2691-1957. doi: 10.1145/3458754.
- Matthias Held and Theresa Velden. How to interpret algorithmically constructed topical structures of scientific fields? a case study of citation-based mappings of the research specialty of invasion biology. *Quantitative Science Studies*, 3(3):651–671, 2022. ISSN 2641-3337. doi: 10.1162/qss_a_00194.
- Nathaniel Imel and Zachary Hafén. Density, asymmetry and citation dynamics in scientific literature. arXiv:2506.23366, 2025. URL <https://arxiv.org/abs/2506.23366>. Preprint.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey Hinton. Similarity of neural network representations revisited. In Kamalika Chaudhuri and Ruslan Salakhutdinov, editors, *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, pages 3519–3529. PMLR, 09–15 Jun 2019. URL <https://proceedings.mlr.press/v97/kornblith19a.html>.
- Jinhyuk Lee, Wonjin Yoon, Sungdong Kim, Donghyeon Kim, Sunkyu Kim, Chan Ho So, and Jaewoo Kang. Biobert: a pre-trained biomedical language representation model for biomedical text mining. *Bioinformatics*, 36(4):1234–1240, 2020. ISSN 1367-4803. doi: 10.1093/bioinformatics/btz682.
- Alex Murphy, Joel Zylberberg, and Alona Fyshe. Correcting biased centered kernel alignment measures in biological and artificial neural networks. arXiv:2405.01012, 2024. URL <https://arxiv.org/abs/2405.01012>. Preprint.
- OpenAlex. Topics. OpenAlex Help Center, 2026. URL <https://help.openalex.org/data/topics/>. Accessed 18 September 2026.

- Malte Ostendorff, Nils Rethmeier, Isabelle Augenstein, Bela Gipp, and Georg Rehm. Neighborhood contrastive learning for scientific document representations with citation embeddings. In Yoav Goldberg, Zornitsa Kozareva, and Yue Zhang, editors, *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*, pages 11670–11688, Abu Dhabi, United Arab Emirates, dec 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.emnlp-main.802. URL <https://aclanthology.org/2022.emnlp-main.802/>.
- Jason Priem, Heather Piwowar, and Richard Orr. OpenAlex: A fully-open index of scholarly works, authors, venues, institutions, and concepts, 2022. URL <https://arxiv.org/abs/2205.01833v2>. Preprint, version 2.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using Siamese BERT-networks. In Kentaro Inui, Jing Jiang, Vincent Ng, and Xiaojun Wan, editors, *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, pages 3982–3992, Hong Kong, China, nov 2019. Association for Computational Linguistics. doi: 10.18653/v1/D19-1410. URL <https://aclanthology.org/D19-1410/>.
- William Rudman, Nate Gillman, Taylor Rayne, and Carsten Eickhoff. IsoScore: Measuring the uniformity of embedding space utilization. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio, editors, *Findings of the Association for Computational Linguistics: ACL 2022*, pages 3325–3339, Dublin, Ireland, may 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.findings-acl.262. URL <https://aclanthology.org/2022.findings-acl.262/>.
- Amanpreet Singh, Mike D’Arcy, Arman Cohan, Doug Downey, and Sergey Feldman. SciRepEval: A multi-format benchmark for scientific document representations. In Houda Bouamor, Juan Pino, and Kalika Bali, editors, *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 5548–5566, Singapore, dec 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.emnlp-main.338. URL <https://aclanthology.org/2023.emnlp-main.338/>.
- Shruti Singh and Mayank Singh. The inefficiency of language models in scholarly retrieval: An experimental walk-through. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio, editors, *Findings of the Association for Computational Linguistics: ACL 2022*, pages 3153–3173, Dublin, Ireland, may 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.findings-acl.249. URL <https://aclanthology.org/2022.findings-acl.249/>.
- Alex H Williams. Equivalence between representational similarity analysis, centered kernel alignment, and canonical correlations analysis. In Marco Fumero, Clementine Domine, Zorah Löhner, Donato Crisostomi, Luca Moschella, and Kimberly Stachenfeld, editors, *Proceedings of UniReps: the Second Edition of the Workshop on Unifying Representations in Neural Models*, volume 285 of *Proceedings of Machine Learning Research*, pages 10–23. PMLR, 14 Dec 2024. URL <https://proceedings.mlr.press/v285/williams24a.html>.
- Alex H Williams, Erin Kunz, Simon Kornblith, and Scott Linderman. Generalized shape metrics on neural representations. In M. Ranzato, A. Beygelzimer, Y. Dauphin, P.S. Liang, and J. Wortman Vaughan, editors, *Advances in Neural Information Processing Systems*, volume 34, pages 4738–4750. Curran Associates, Inc., 2021. URL https://proceedings.neurips.cc/paper_files/paper/2021/file/252a3dbaeb32e7690242ad3b556e626b-Paper.pdf.